(Please write your Enrollment Number)

Enrollment No. 02701022022

MID-TERM EXAMINATION
(Course Name: B.Tech ECE) (Semester 3rd)
October, 2023

Subject Code: BEC 207	**Subject: Digital Electronics**
Time :1 ½Hours	**Maximum Marks : 30**
Note: All Q. is Compulsory.	

Q1	**UNIT - 1**	**(10)**	
	(a) Differentiate between (with example) (i) Weighted Code and Non-Weighted Code (ii) Reflective Code and Self-Complementing Code		(4)
	(b) Prove the following Boolean Expressions $(A + B)(A + C) = A + BC$		(2)
	(c) Draw the Internal Circuitry for RS flip flop using only NAND gates. State its truth table too.		(3)
	(d) Covert the Gray code 1001011 to binary		(1)
Q2	**UNIT - 2**	**(10)**	
	(a) Implement in NOR logic. $Y = (A + B + C)(D + E)\bar{F}$		(2)
	(b) Convert the expression into canonical SOP form $Y = AB + A\bar{C} + BC$		(2)
	(c) Find the minimum POS and SOP expressions for the function using k – maps $F(ABCD) = \prod M(0, 2, 6, 8, 9, 10).\ D(5, 12, 13, 14, 15)$		(6)
Q3	**UNIT - 3**	**(10)**	
	(a) The 74154 is a 4:16 decoder with active low outputs. Using this IC and logic gates, implement the following Boolean functions. $F^1 = \sum m(1, 9, 12, 15)$ $F_2 = \sum m(2, 5, 9)$		(4)
	(b) Implement a 16 : 1 MUX using 4 : 1 MUX		(3)
	(c) What is race around condition in flip flops. How can it be corrected.		(3)